

```
"""
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下載真實台股日K，輸出 data.json 給網頁使用。

GitHub Actions 每個交易日收盤後自動執行；也可以在自己電腦跑：python update_data.py

```
"""
```

```
import json, sys, datetime as dt
```

```
import pandas as pd
```

```
import yfinance as yf
```

```
TECH = ["2330", "2454", "2317", "2382", "3231", "6669", "2356", "2376", "2357", "  
        "2308", "2345", "3037", "8046", "3189", "2379", "3034", "3661", "3443", "  
        "2327", "2383", "6274", "3529", "2449", "4966"]
```

```
MIXED = ["2330", "2317", "2454", "2308", "2382", "2303", "2881", "2882", "2891",  
        "1301", "2002", "2603", "2609", "2615", "3231", "2357", "2376", "3037",  
        "2379", "3008", "2395", "1216", "2207", "5871"]
```

```
BENCH = "0050"
```

```
OTC = {"3324", "6274", "3529"} # 上櫃股票在 Yahoo 用 .TWO
```

```
def sym(code, otc=None):
```

```
    return code + (".TWO" if (code in OTC if otc is None else otc) else ".TW")
```

```
def to_dict(df):
```

```
    df = df.dropna()
```

```
    df = df[df["Volume"] > 0]
```

```
    r = lambda s: [round(float(x), 2) for x in s]
```

```
    return dict(dates=[d.strftime("%Y-%m-%d") for d in df.index], o=r(df["Open"])  
                l=r(df["Low"]), c=r(df["Close"]), v=[int(x) for x in df["Volume"]])
```

```
def download(symbols):
```

```
    raw = yf.download(symbols, period="3y", group_by="ticker", auto_adjust=True,  
                      out = {})
```

```
    for s in symbols:
```

```
        try:
```

```
            df = raw[s] if len(symbols) > 1 else raw
```

```
            if isinstance(df.columns, pd.MultiIndex):
```

```
                df = df.droplevel(0, axis=1) if df.columns.nlevels > 1 else df
```

```
            d = to_dict(df)
```

```
            if len(d["dates"]) >= 60:
```

```
                out[s] = d
```

```
        except Exception:
```

```
            pass
```

```
    return out
```

```

def main():
    codes = sorted(set(TECH + MIXED)) + [BENCH]
    got = download([sym(c) for c in codes])
    stocks = {c: got[sym(c)] for c in codes if sym(c) in got}
    missing = [c for c in codes if c not in stocks]
    if missing: # 換另一個市場代號再試一次
        alt = download([sym(c, otc=c not in OTC) for c in missing])
        for c in missing:
            if sym(c, otc=c not in OTC) in alt:
                stocks[c] = alt[sym(c, otc=c not in OTC)]
    if BENCH not in stocks or len(stocks) < 20:
        sys.exit(f"下載失敗，只拿到 {len(stocks)} 檔，不更新 data.json")
    bench = stocks.pop(BENCH)
    now = dt.datetime.now(dt.timezone(dt.timedelta(hours=8)))
    data = dict(stocks=stocks, bench=dict(dates=bench["dates"], c=bench["c"]), so
                updated=now.strftime("%Y-%m-%d %H:%M"), last_date=bench["dates"][
    with open("data.json", "w", encoding="utf-8") as f:
        json.dump(data, f, ensure_ascii=False, separators=(",", ":"))
    print(f"完成：{len(stocks)} 檔，最新交易日 {data['last_date']}", 缺少 {[c for c in

if __name__ == "__main__":
    main()

```